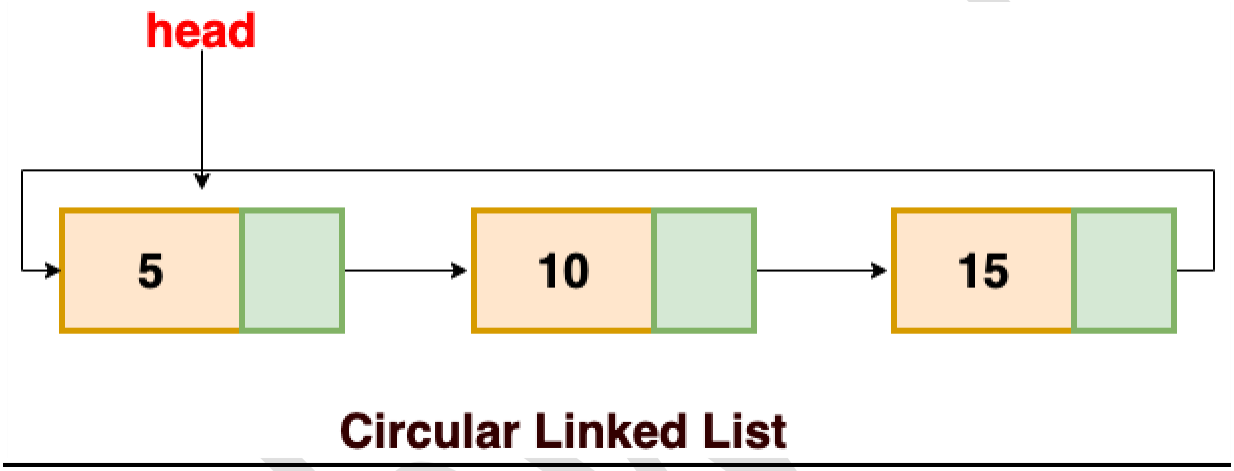
Circular Linked List

Circular Linked List: A circular linked list is either a singly or doubly linked list in which there are no **NULL** values. Here, we can implement the Circular Linked List by making the use of Singly or Doubly Linked List. In the case of a singly linked list, the next of the last node contains the address of the first node and in case of a doubly-linked list, the next of last node contains the address of the first node and prev of the first node contains the address of the last node.



Advantages of a Circular linked list

- Entire list can be traversed from any node.
- Circular lists are the required data structure when we want a list to be accessed in a circle or loop.
- Despite of being singly circular linked list we can easily traverse to its previous node, which is not possible in singly linked list.

Disadvantages of Circular linked list

- Circular list are complex as compared to singly linked lists.
- Reversing of circular list is a complex as compared to singly or doubly lists.
- If not traversed carefully, then we could end up in an infinite loop.

- Like singly and doubly lists circular linked lists also doesn't supports direct accessing of elements.

Applications of Circular linked list

- Circular lists are used in applications where the entire list is accessed one-by-one in a loop. Example: Operating systems may use it to switch between various running applications in a circular loop.
- It is also used by Operating system to share time for different users, generally uses Round-Robin time sharing mechanism.
- Multiplayer games uses circular list to swap between players in a loop.

C program to create and traverse Circular Linked List

```
#include  <stdio.h>

#include <stdlib.h>

struct node {

   int data;

   struct node * next;

}*head;

void createList(int n);

void displayList();

void main()

{

   int n, data, choice=1;
```

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```
    head = NULL;
    while(choice != 0)
    {
        printf("CIRCULAR LINKED LIST PROGRAM\n");
        printf("1. Create List\n");
       printf("2. Display list\n");
       printf("0. Exit\n");
       printf("Enter your choice : ");
scanf("%d", &choice);


        switch(choice)
        {
            case 1:
                printf("Enter the total number of nodes in list: ");
                scanf("%d", &n);
                createList(n);
                break;
            case 2:
                displayList();
                break;
            case 0:
```

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```
            break;
        default:
            printf("Error! Invalid choice. Please choose between 0-2");
    }


    printf("\n\n\n\n\n");
  }
getch();
}
void createList(int n)
{
  int i, data;
  struct node *prevNode, *newNode;


  if(n >= 1)
  {
    head = (struct node *)malloc(sizeof(struct node));


    printf("Enter data of 1 node: ");
    scanf("%d", &data);
```

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```
  head->data = data;
  head->next = NULL;

  prevNode = head;

for(i=2; i<=n; i++)
  {
     newNode = (struct node *)malloc(sizeof(struct node));

     printf("Enter data of %d node: ", i);
     scanf("%d", &data);

     newNode->data = data;
     newNode->next = NULL;
     prevNode->next = newNode;
     prevNode = newNode;
  }

  prevNode->next = head;

  printf("\nCIRCULAR LINKED LIST CREATED SUCCESSFULLY\n");
```

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```
    }
}

void displayList()
{
    struct node *current;
    int n = 1;

    if(head == NULL)
    {
        printf("List is empty.\n");
    }
    else
    {
        current = head;
        printf("DATA IN THE LIST:\n");

        do {
            printf("Data %d = %d\n", n, current->data);

            current = current->next;
```

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```
        n++;
      }while(current != head);
   }
}
```

C program to insert a new node in circular linked list

```
#include  <stdio.h>
#include <stdlib.h>
struct node {
   int data;
   struct node * next;
}*head;
void createList(int n);
void displayList();
void insertAtBeginning(int data);
void insertAtN(int data, int position);
void main()
{
   int n, data, choice=1;


   head = NULL;
```

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```
while(choice != 0)
{
        printf("CIRCULAR LINKED LIST PROGRAM\n");
   printf("1. Create List\n");
   printf("2. Display list\n");
   printf("3. Insert at beginning\n");
   printf("4. Insert at any position\n");
   printf("0. Exit\n");
   printf("Enter your choice : ");

   scanf("%d", &choice);

   switch(choice)
   {
      case 1:
         printf("Enter the total number of nodes in list: ");
         scanf("%d", &n);
         createList(n);
         break;
```

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```
    case 2:
        displayList();
        break;
    case 3:
        printf("Enter data to be inserted at beginning: ");
        scanf("%d", &data);
        insertAtBeginning(data);
        break;
    case 4:
        printf("Enter node position: ");
        scanf("%d", &n);
        printf("Enter data you want to insert at %d position: ", n);
        scanf("%d", &data);
        insertAtN(data, n);
        break;
    case 0:
        break;
    default:
        printf("Error! Invalid choice. Please choose between 0-4");
}
```

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```
        printf("\n\n\n\n\n");
    }

getch();
}
void createList(int n)
{
    int i, data;
    struct node *prevNode, *newNode;

    if(n >= 1)
    {
        head = (struct node *)malloc(sizeof(struct node));

        printf("Enter data of 1 node: ");
        scanf("%d", &data);

        head->data = data;
        head->next = NULL;

        prevNode = head;
```

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```
    for(i=2; i<=n; i++)
    {
        newNode = (struct node *)malloc(sizeof(struct node));

        printf("Enter data of %d node: ", i);
        scanf("%d", &data);

        newNode->data = data;
        newNode->next = NULL;
        prevNode->next = newNode;

        prevNode = newNode;
    }
    prevNode->next = head;

    printf("\nCIRCULAR LINKED LIST CREATED SUCCESSFULLY\n");
  }
}
```

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```
void displayList()
{
    struct node *current;
    int n = 1;

    if(head == NULL)
    {
        printf("List is empty.\n");
    }
    else
    {
        current = head;
        printf("DATA IN THE LIST:\n");

        do {
            printf("Data %d = %d\n", n, current->data);

            current = current->next;
            n++;
        }while(current != head);
    }
```

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```
}


void insertAtBeginning(int data)
{
    struct node *newNode, *current;


    if(head == NULL)
    {
        printf("List is empty.\n");
    }
    else
    {
        newNode = (struct node *)malloc(sizeof(struct node));
        newNode->data = data;
        new current = head;
        while(current->next != head)
        {
            current = current->next;
        }
        current->next = newNode;
```

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```
            head = newNode;


        printf("NODE INSERTED SUCCESSFULLY\n");
    }
}
void insertAtN(int data, int position)
{
    struct node *newNode, *current;
    int i;


    if(head == NULL)
    {
        printf("List is empty.\n");
    }
    else if(position == 1)
    {
        insertAtBeginning(data);
    }
    else
    {
```

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```
        newNode = (struct node *)malloc(sizeof(struct node));

        newNode->data = data;

        current = head;

        for(i=2; i<=position-1; i++)

        {

            current = current->next;

        }

        newNode->next = current->next;

        current->next = newNode;



        printf("NODE INSERTED SUCCESSFULLY.\n");

    }

}
```

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